Shikhar Saxena

saxena.shikhar1808@gmail.com | +91 87550 43412

linkedin.com/in/shikhar-saxena-90a484296 | github.com/Shikhar1808 | leetcode.com/u/ShikharSaxena6251

Education

Thapar Institute of Engineering and Technology, Patiala Aug 2023 – Jun 2027 (Expected)
- B.E. Computer Science and Engineering | CGPA: 8.59/10

Army Public School, Bareilly Class of 2023
- **12th:** 92.4%, **10th:** 93%

Experience

Voxomos Systems Pvt Ltd, Software Intern Jun 2026 – Jul 2026
- Developed and benchmarked CPU, **CUDA**, and **HIP** implementations of cosine similarity and vector search workloads using NVIDIA Nsight Systems and Nsight Compute, exploring cross-platform GPU portability via **AMD ROCm**, **HIP**, and **HIPIFY**
- Evaluated inference performance across **vLLM**, **SGLang**, and **TensorRT-LLM**, isolating KV-cache memory limits under load; sustained **14–15 ms/token decode latency** and mitigated Time-to-First-Token (**TTFT**) spikes by optimizing request concurrency.

TruckHai, Backend Intern Jan 2026 – Apr 2026
- Built Go (Gin, GORM) **REST microservices** for fleet dispatch, modelling the trip lifecycle as a **finite-state machine** with **ACID transactions** so that invalid state transitions are rejected and concurrent updates stay consistent.
- Removed race conditions and row-lock contention in **PostgreSQL** through **composite indexing** and query-plan tuning, and added **RBAC middleware** with audit-logged admin impersonation across **5+ user roles**.

Projects

Relay |*Python, FastAPI, MCP, LangGraph, PostgreSQL, Redis, Kubernetes, Helm, Next.js* **github.com/Shikhar1808/Relay**
- Built an **MCP gateway** that combines **11 tools from 3 MCP servers** I wrote behind one **Streamable HTTP endpoint**, with per-server timeouts so one dead server never breaks the rest. Added a LangGraph chat agent that acts as the signed-in user and asks for confirmation before any write.
- Secured it with **SHA-256-hashed API keys, AES-256-GCM-encrypted credentials** and a hash-chained audit log, and deployed it on Kubernetes with Helm and **deny-by-default NetworkPolicies.**

ZaiGuard — Surveillance System | *FastAPI, PostgreSQL, Qdrant* **github.com/Shikhar1808/Zaiguard-Connected**
- Built a 5-layer **FastAPI** middleware (Redis TTL sliding windows, PostgreSQL B-Trees, Qdrant HNSW vector suppression) that reduced false-positive alarms by **65%–90%** in testing at sub-second latency.
- Used a **Transactional Outbox** worker to prevent PostgreSQL/Qdrant dual-write inconsistencies; coordinated cross-device **Flower federated learning** across LAN SuperNodes.
- Built edge video ingestion with motion-gated frame sampling and an RTSP/HTTP/USB fallback chain, feeding an **ONNX**-accelerated **YOLO** detector (**96.5% precision**) with **ByteTrack**.

Zenith Website | *NodeJS, ExpressJS, MongoDB, JWT Auth, NGINX, AWS* **github.com/Shikhar1808/ZenithCryptic**
- Built a competition platform serving **1000+ concurrent users** at **sub-100 ms** response times; designed real-time leaderboard calculation and stateless **JWT** auth to remove database bottlenecks during submission spikes.
- Secured the deployment with **Cloudflare** DDoS mitigation and **NGINX** reverse-proxy rate limiting and load balancing.

Technologies and Skills

Languages: C++, C, Go, Python, JavaScript, TypeScript, SQL, Solidity
Backend & Web: Microservices, REST APIs, Node.js, Express.js, FastAPI, Gin, GORM, React.js, Tailwind CSS, Socket.io, JWT, RBAC
AI & GPU Computing: PyTorch, CUDA, AMD ROCm/HIP, vLLM, LangChain, LangGraph, MCP, OpenCV, Qdrant, Flower
Databases & DevOps: PostgreSQL, MongoDB, MySQL, Redis, Docker, Linux, GitHub Actions (CI/CD), NGINX, Prometheus, Grafana, Git, AWS, Kubernetes, Helm

Awards and Certifications

SPHNIX'24 Hackathon – 1st Runner Up, MNIT Jaipur Nov 2024
NVIDIA Certified: Accelerated Computing (Modern CUDA C++, CUDA Python), AI on Jetson Nano, Generative AI & Deep Learning 2025 – 2026
LeetCode: 417+ problems solved, 1551 Contest Rating (DSA: arrays, graphs, DP, trees, stacks, queues, heap)